

```
In [1]: import pandas as pd
```

```
In [2]: pd.read_csv(r"C:\Users\user\Desktop\Dataset\supermarket_sales.csv")
```

```
Out[2]:
```

	InvoiceID	Date	Branch	Product	Quantity	UnitPrice	TotalAmount
0	INV00001	2025-05-21	Nakuru	Milk	1	80	80
1	INV00002	2025-02-14	Eldoret	Milk	9	80	720
2	INV00003	2025-04-30	Eldoret	Milk	4	80	320
3	INV00004	2025-08-03	Nakuru	Rice	9	200	1800
4	INV00005	2025-06-24	Kisumu	Sugar	7	120	840
...	...	...	...	...	...	...	...
1495	INV01496	2025-07-21	Nakuru	Bread	10	50	500
1496	INV01497	2025-07-02	Nairobi	Juice	1	220	220
1497	INV01498	2025-02-24	Nairobi	Cooking Oil	6	350	2100
1498	INV01499	2025-02-21	Mombasa	Soap	1	150	150
1499	INV01500	2025-03-05	Mombasa	Sugar	7	120	840

1500 rows × 10 columns

```
In [3]: df=pd.read_csv(r"C:\Users\user\Desktop\Dataset\supermarket_sales.csv")
```

```
In [4]: df
```

Out[4]:

	InvoiceID	Date	Branch	Product	Quantity	UnitPrice	TotalAmou
0	INV00001	2025-05-21	Nakuru	Milk	1	80	80
1	INV00002	2025-02-14	Eldoret	Milk	9	80	720
2	INV00003	2025-04-30	Eldoret	Milk	4	80	320
3	INV00004	2025-08-03	Nakuru	Rice	9	200	1800
4	INV00005	2025-06-24	Kisumu	Sugar	7	120	840
...	...	...	...	...	...	...	...
1495	INV01496	2025-07-21	Nakuru	Bread	10	50	500
1496	INV01497	2025-07-02	Nairobi	Juice	1	220	220
1497	INV01498	2025-02-24	Nairobi	Cooking Oil	6	350	2100
1498	INV01499	2025-02-21	Mombasa	Soap	1	150	150
1499	INV01500	2025-03-05	Mombasa	Sugar	7	120	840

1500 rows × 10 columns

In [5]: `df.shape`

Out[5]: (1500, 10)

In [6]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   InvoiceID              1500 non-null   object
1   Date                  1500 non-null   object
2   Branch                1500 non-null   object
3   Product               1500 non-null   object
4   Quantity              1500 non-null   int64
5   UnitPrice             1500 non-null   int64
6   TotalAmount           1500 non-null   int64
7   PaymentMethod         1500 non-null   object
8   CustomerGender        1500 non-null   object
9   CustomerAge           1500 non-null   int64
dtypes: int64(4), object(6)
memory usage: 117.3+ KB
```

In [7]: `df.dtypes`

```
Out[7]: InvoiceID      object
Date      object
Branch    object
Product   object
Quantity  int64
UnitPrice int64
TotalAmount int64
PaymentMethod object
CustomerGender object
CustomerAge int64
dtype: object
```

```
In [8]: #missing values
df.isnull().sum()
```

```
Out[8]: InvoiceID      0
Date      0
Branch    0
Product   0
Quantity  0
UnitPrice  0
TotalAmount 0
PaymentMethod 0
CustomerGender 0
CustomerAge 0
dtype: int64
```

```
In [9]: df.duplicated().sum()
```

```
Out[9]: np.int64(0)
```

```
In [10]: df.describe(include='all')
```

```
Out[10]:
```

	InvoiceID	Date	Branch	Product	Quantity	UnitPrice	TotalAmount
count	1500	1500	1500	1500	1500.000000	1500.000000	1500.000000
unique	1500	357	5	8	NaN	NaN	NaN
top	INV00001	2025-02-28	Mombasa	Juice	NaN	NaN	NaN
freq	1	14	317	205	NaN	NaN	NaN
mean	NaN	NaN	NaN	NaN	5.548000	168.566667	924.840000
std	NaN	NaN	NaN	NaN	2.855369	86.152735	704.152735
min	NaN	NaN	NaN	NaN	1.000000	50.000000	50.000000
25%	NaN	NaN	NaN	NaN	3.000000	120.000000	360.000000
50%	NaN	NaN	NaN	NaN	6.000000	150.000000	720.000000
75%	NaN	NaN	NaN	NaN	8.000000	220.000000	1320.000000
max	NaN	NaN	NaN	NaN	10.000000	350.000000	3500.000000

```
In [11]: #categories
df.PaymentMethod.value_counts()
```

```
Out[11]: PaymentMethod
M-Pesa    506
Cash      503
Card      491
Name: count, dtype: int64
```

```
In [12]: #total revenue
df.TotalAmount.sum()
```

```
Out[12]: np.int64(1402420)
```

```
In [13]: #most sold product
df.Product.mode()
```

```
Out[13]: 0    Juice
Name: Product, dtype: object
```

```
In [14]: #branch with the highest revenue
#total branches
df.Branch.value_counts()
```

```
Out[14]: Branch
Mombasa    317
Nakuru     309
Nairobi    296
Eldoret    295
Kisumu     283
Name: count, dtype: int64
```

```
In [15]: df.Branch.dtypes
```

```
Out[15]: dtype('O')
```

```
In [16]: #branch generated the highest revenue
df.groupby('Branch')['TotalAmount'].sum().sort_values(ascending=True)
```

```
Out[16]: Branch
Kisumu     267790
Nakuru     271350
Nairobi    271600
Eldoret    294540
Mombasa    297140
Name: TotalAmount, dtype: int64
```

```
In [17]: #average customer age
df.CustomerAge.mean()
```

```
Out[17]: np.float64(41.916)
```

```
In [18]: df.head()
```

```
Out[18]:
```

	InvoiceID	Date	Branch	Product	Quantity	UnitPrice	TotalAmount	Pi
0	INV00001	2025-05-21	Nakuru	Milk	1	80	80	
1	INV00002	2025-02-14	Eldoret	Milk	9	80	720	
2	INV00003	2025-04-30	Eldoret	Milk	4	80	320	
3	INV00004	2025-08-03	Nakuru	Rice	9	200	1800	
4	INV00005	2025-06-24	Kisumu	Sugar	7	120	840	

```
In [19]: #gender shopped the most
#groups of gender
df.CustomerGender.value_counts()
```

```
Out[19]: CustomerGender
Female    753
Male      747
Name: count, dtype: int64
```

```
In [20]: df.groupby('CustomerGender')['TotalAmount'].sum()
```

```
Out[20]: CustomerGender
Female    702750
Male      699670
Name: TotalAmount, dtype: int64
```

```
In [21]: #product that brought more revenue
df.groupby('Product')['TotalAmount'].sum().sort_values(ascending=True)
```

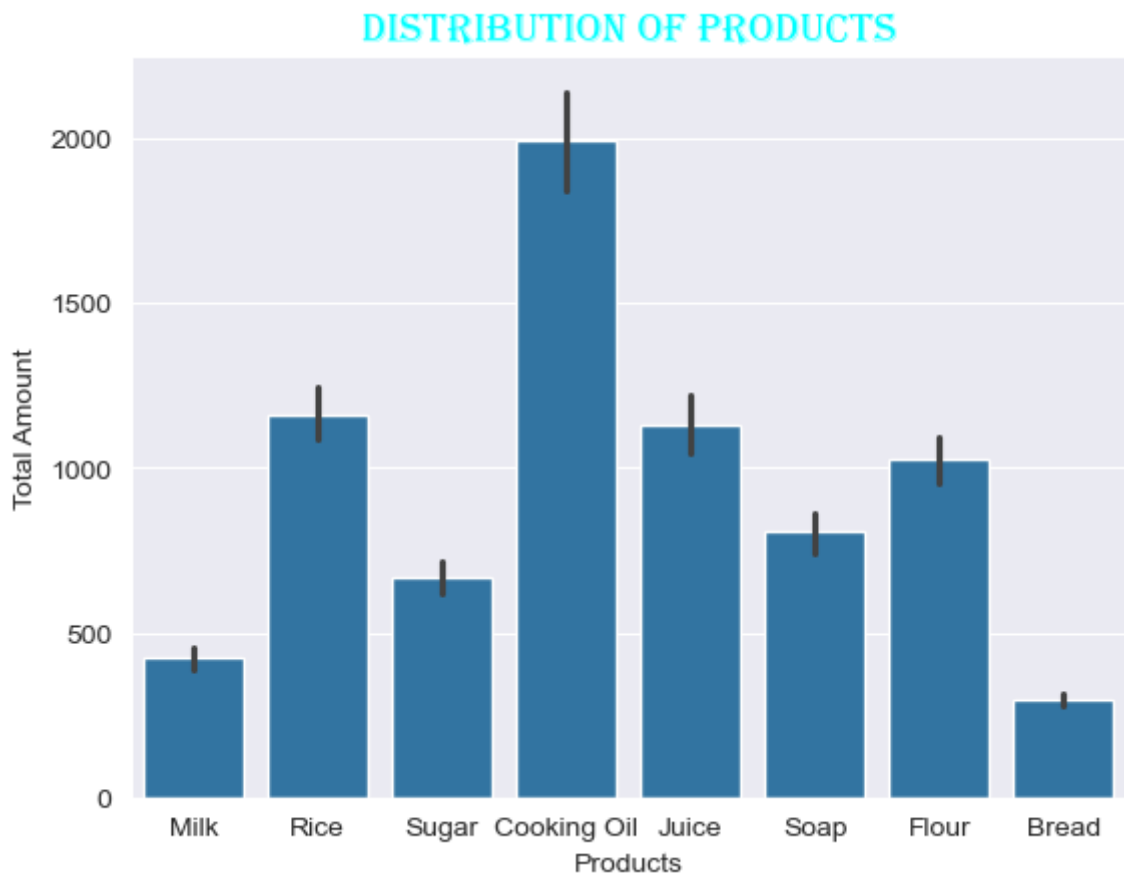
```
Out[21]: Product
Bread          51500
Milk           81440
Sugar         131640
Soap          151350
Flour         189000
Rice          209400
Juice         231440
Cooking Oil   356650
Name: TotalAmount, dtype: int64
```

```
In [22]: df.groupby('Product')['TotalAmount'].sum().sort_values(ascending=False)
```

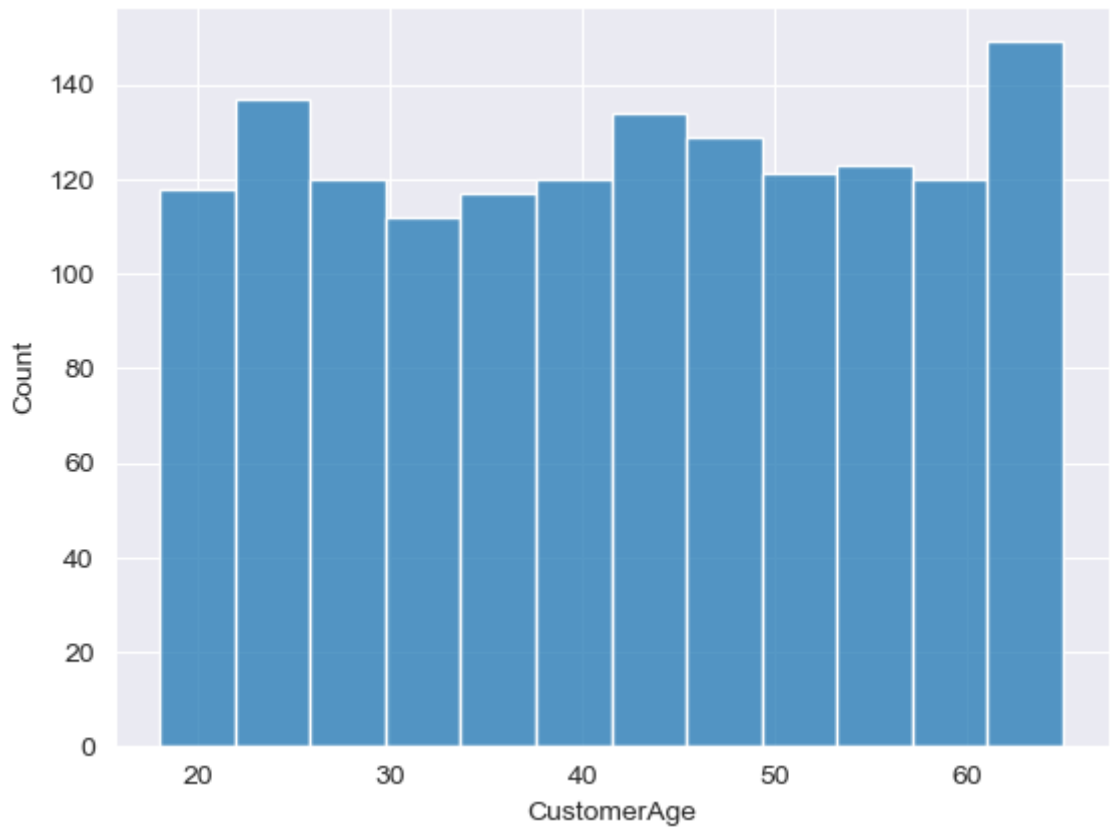
```
Out[22]: Product
Cooking Oil   356650
Juice        231440
Rice         209400
Flour        189000
Soap         151350
Sugar        131640
Milk         81440
Bread        51500
Name: TotalAmount, dtype: int64
```

```
In [23]: import seaborn as sns
```

```
In [32]: #plot chart to visualize most sold product
from matplotlib import pyplot as plt
sns.barplot(x='Product',y='TotalAmount',data=df)
plt.title('Distribution of Products',fontname='Algerian',fontsize=15,c='c')
plt.xlabel('Products',fontname='arial')
plt.ylabel('Total Amount')
plt.savefig("graph.png")
plt.show()
#modify fontsize
```



```
In [34]: #Visualize distribution of customer age
sns.histplot(x='CustomerAge',data=df)
plt.savefig("graph.png")
plt.show()
```

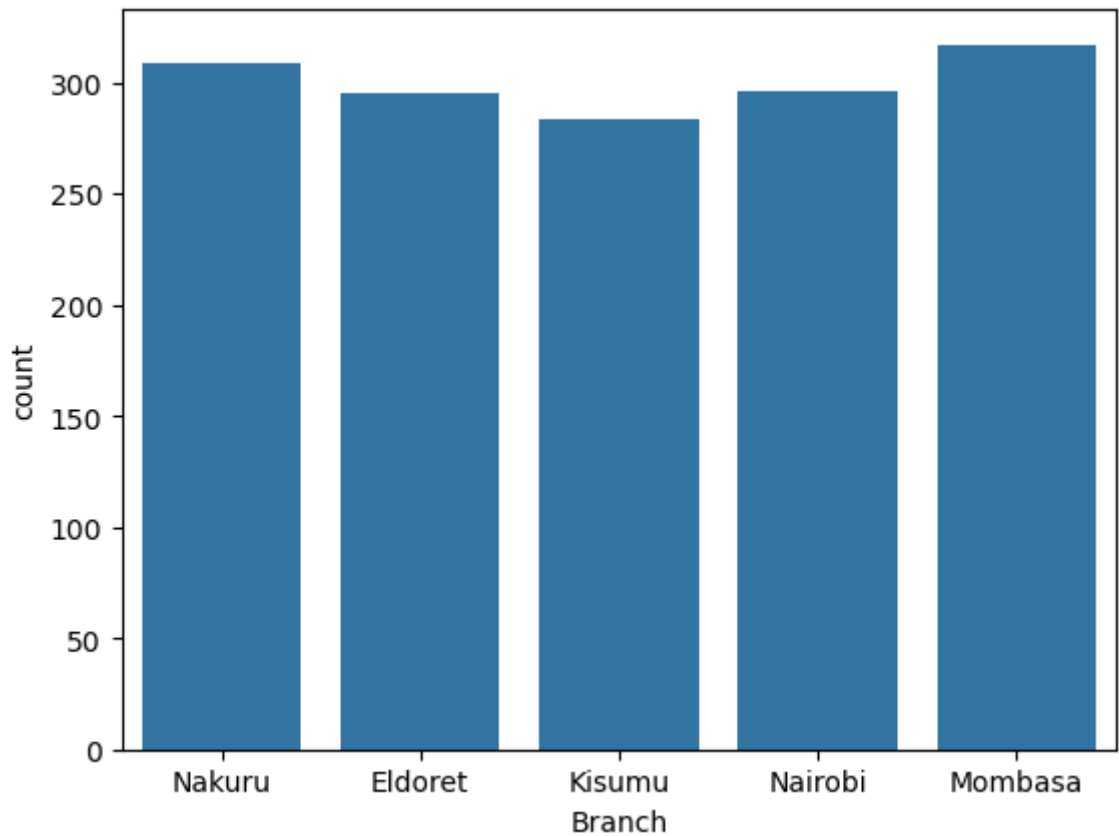


```
In [26]: #check version of software
import sys
print (sys.version)
```

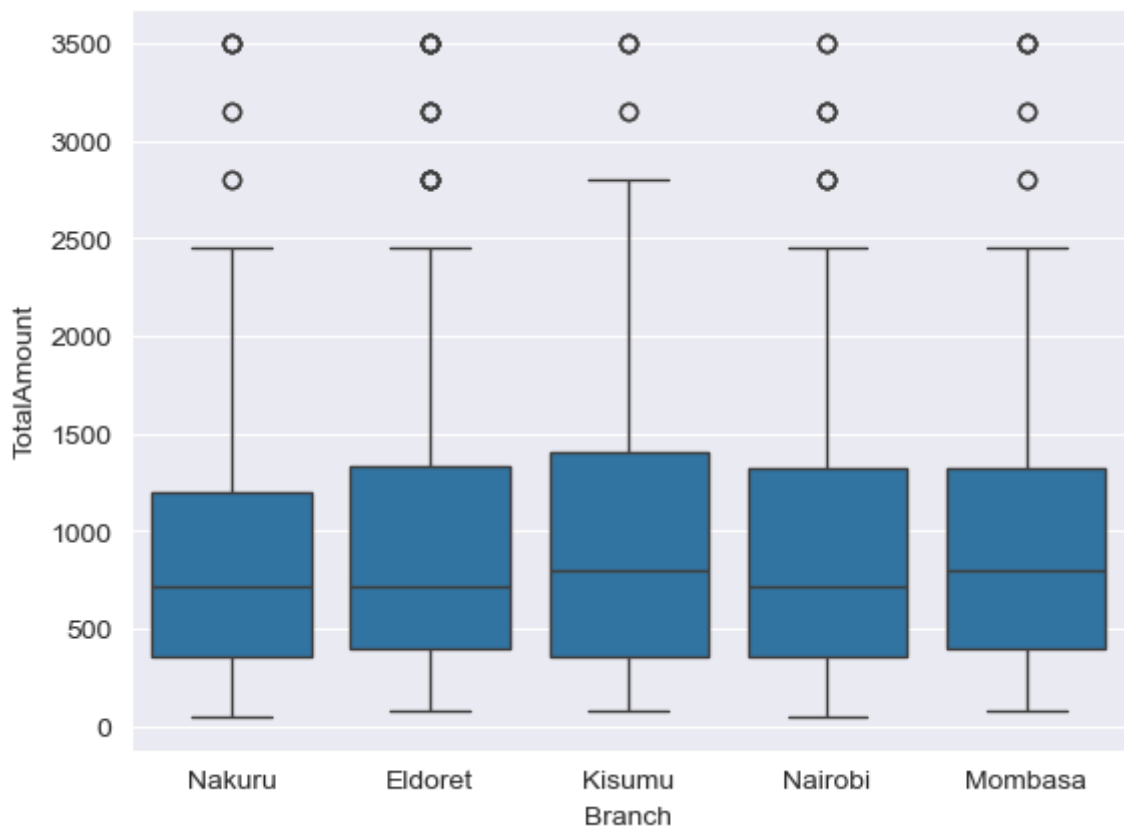
3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1929 64 bit (AMD64)]

```
In [27]: #branch with highest number of customers
sns.countplot(x='Branch',data=df)
```

Out[27]: <Axes: xlabel='Branch', ylabel='count'>



```
In [35]: #Branch with highest median revenue?
sns.boxplot(x='Branch',y='TotalAmount',data=df)
sns.set_style('darkgrid')
plt.savefig("graph.png")
#kisumu is the highest
```



```
In [ ]:
```



```
In [2]: import pandas as pd
```

```
In [3]: df=pd.read_excel(r"C:\Users\user\Downloads\students_dataset.xlsx")
```

```
In [4]: df
```

```
Out[4]:
```

	stud_id	age_yrs	sex	course_taken	yr_of_study	att_perc	study
0	STU1000	23.0	Male	Statistics	1	79.5	
1	STU1001	31.0	F	Statistics	3	94.8	
2	STU1002	27.0	female	Computer Science	2	67.9	
3	STU1003	24.0	male	Applied Maths	4	92.4	
4	STU1004	23.0	Male	Computer Science	4	47.1	
...	...	...	...	...	...	...	...
1220	STU2006	23.0	Male	Statistics	3	69.1	
1221	STU1306	24.0	F	Computer Science	2	94.5	
1222	STU2092	29.0	female	Data Science	3	75.4	
1223	STU1568	26.0	male	Applied Maths	1	80.3	
1224	STU1220	19.0	F	IT	1	80.6	

1225 rows × 14 columns

```
In [5]: #dimensions
#14 variables
#1225 observations
```

```
In [6]: #categorical variables& numericals
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1225 entries, 0 to 1224
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   stud_id               1217 non-null   object
1   age_yrs               1200 non-null   float64
2   sex                  1225 non-null   object
3   course_taken         1225 non-null   object
4   yr_of_study          1225 non-null   int64
5   att_perc             1225 non-null   float64
6   study_hrs_week       1225 non-null   float64
7   assgn_score          1198 non-null   float64
8   final_exam          1200 non-null   float64
9   fam_income_kes       1200 non-null   float64
10  transport_mode        1225 non-null   object
11  phone_no              1225 non-null   int64
12  lecturer_notes       1019 non-null   object
13  temp_var_delete      1225 non-null   int64
dtypes: float64(6), int64(3), object(5)
memory usage: 134.1+ KB
```

```
In [7]: df.nunique()
```

```
Out[7]: stud_id      1192
age_yrs      24
sex          6
course_taken  6
yr_of_study  4
att_perc     453
study_hrs_week 305
assgn_score  530
final_exam  579
fam_income_kes 1170
transport_mode 7
phone_no     1200
lecturer_notes 5
temp_var_delete 99
dtype: int64
```

```
In [8]: df.course_taken.value_counts()
```

```
Out[8]: course_taken
Data Science      214
Applied Maths     209
Economics         209
Statistics        207
IT                204
Computer Science  182
Name: count, dtype: int64
```

```
In [9]: df.sex.value_counts()
```

```
Out[9]: sex
female      218
F           216
Female      212
Male        202
male        197
M           180
Name: count, dtype: int64
```

```
In [10]: #Standardised use replace function
df.sex.replace({'male':'Male','female':'Female','M':'Male','F':'Female'},
#methods to parmanents change
#inplce=True
#or df.sex=df.sexreplace({'male':'Male','female':'Female','M':'Male','F':
```

C:\Users\user\AppData\Local\Temp\ipykernel_7900\2834635892.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df.sex.replace({'male': 'Male', 'female': 'Female', 'M': 'Male', 'F': 'Female'}, inplace=True)
```

```
In [11]: #percentage of male and female  
df.sex.value_counts(normalize=True)*100
```

```
Out[11]: sex  
Female    52.734694  
Male      47.265306  
Name: proportion, dtype: float64
```

```
In [12]: #Which course was taken by higher no. of students  
df.course_taken.value_counts(normalize=True)*100
```

```
Out[12]: course_taken  
Data Science    17.469388  
Applied Maths   17.061224  
Economics       17.061224  
Statistics      16.897959  
IT              16.653061  
Computer Science 14.857143  
Name: proportion, dtype: float64
```

```
In [13]: #unique values in lecture notes  
df.lecturer_notes.value_counts()
```

```
Out[13]: lecturer_notes  
Good           225  
Late           214  
Excellent      203  
Needs follow-up 190  
-              187  
Name: count, dtype: int64
```

```
In [14]: df.transport_mode.value_counts(normalize=True)*100
```

```
Out[14]: transport_mode  
Walk          17.795918  
Private        17.224490  
bus           16.897959  
Boda           16.408163  
Matatu         15.346939  
Bus            14.693878  
MATATU         1.632653  
Name: proportion, dtype: float64
```

```
In [75]: df.transport_mode.replace({'bus': 'Bus', 'MATATU': 'Matatu'}, inplace=True)
```

```
In [80]: df.transport_mode.replace({'MATATU': 'Matatu', 'Bus': 'Matatu'}, inplace=True)
```

```
In [82]: df.transport_mode.replace({'matatu': 'Matatu'}, inplace=True)
```

```
In [86]: df.isnull().sum()
```

```
Out[86]: stud_id      8
         age_yrs    25
         sex        0
         course_taken  0
         yr_of_study  0
         att_perc    0
         study_hrs_week  0
         assgn_score  27
         final_exam  25
         fam_income_kes  25
         transport_mode  0
         phone_no     0
         lecturer_notes  206
         temp_var_delete  0
         dtype: int64
```

```
In [92]: df.duplicated().sum()
```

```
Out[92]: np.int64(23)
```

```
In [93]: #display the duplicates we filter
         df[df.duplicated()]
```

Out[93]:	stud_id	age_yrs	sex	course_taken	yr_of_study	att_perc	study
1200	STU1636	35.0	Male	IT	2	76.2	
1201	STU1683	33.0	Male	Economics	3	65.3	
1204	STU2083	33.0	Female	Statistics	2	69.3	
1205	STU1283	19.0	Female	Data Science	2	92.7	
1206	STU1817	34.0	Female	Statistics	4	83.2	
1207	STU1314	18.0	Male	Statistics	2	74.7	
1208	STU1537	17.0	Male	Statistics	1	82.5	
1209	STU1664	24.0	Female	IT	3	85.0	
1210	STU1239	17.0	Female	Computer Science	2	73.2	
1211	STU1135	27.0	Male	Economics	2	58.7	
1212	STU1335	29.0	Male	Statistics	4	98.5	
1213	STU1189	35.0	Male	Economics	4	94.9	
1214	STU1838	29.0	Female	Economics	1	43.9	
1215	STU1321	21.0	Female	Computer Science	1	100.0	
1216	STU2146	35.0	Male	Statistics	2	86.1	
1217	STU1228	19.0	Female	Computer Science	4	57.2	
1218	STU1480	30.0	Female	Data Science	1	83.9	
1219	STU2043	26.0	Female	IT	3	90.6	
1220	STU2006	23.0	Male	Statistics	3	69.1	
1221	STU1306	24.0	Female	Computer Science	2	94.5	
1222	STU2092	29.0	Female	Data Science	3	75.4	
1223	STU1568	26.0	Male	Applied Maths	1	80.3	
1224	STU1220	19.0	Female	IT	1	80.6	

```
In [108... #delete duplicates
#first check for unique values in students id
df.stud_id.value_counts()
```

```
Out[108...] stud_id
STU2043      2
STU1283      2
STU1135      2
STU1636      2
STU1568      2
..
STU1404      1
STU1403      1
STU1402      1
STU1401      1
STU2199      1
Name: count, Length: 1192, dtype: int64
```

```
In [109...] df.drop_duplicates()
```

```
Out[109...]      stud_id  age_yrs    sex  course_taken  yr_of_study  att_perc  study
0  STU1000    23.0    Male    Statistics           1        79.5
1  STU1001    31.0  Female    Statistics           3        94.8
2  STU1002    27.0  Female  Computer Science           2        67.9
3  STU1003    24.0    Male  Applied Maths           4        92.4
4  STU1004    23.0    Male  Computer Science           4        47.1
...      ...      ...      ...      ...      ...      ...
1197 STU2197    24.0  Female    Statistics           1       100.0
1198 STU2198    28.0    Male    Economics           3        90.4
1199 STU2199    23.0    Male    Economics           4        69.6
1202 STU2033    35.0    Male  Applied Maths           2        69.7
1203 STU1190    19.0  Female    Economics           1        78.1
```

1202 rows × 14 columns

```
In [116...] df[df.stud_id=='STU2043']
```

```
Out[116...]      stud_id  age_yrs    sex  course_taken  yr_of_study  att_perc  study
1043 STU2043    26.0  Female           IT           3        90.6
```

```
In [115...] #delete the above row
df.drop(1219,axis=0,inplace=True)
```

```
In [121...] #reame yr of study into study year
df.rename(columns={'yr_of_study':'Study_Year'},inplace=True)
```

```
In [1]: df.head()
```

```
-----  
-  
NameError                                Traceback (most recent call las  
t)  
Cell In[1], line 1  
----> 1 df.head()  
  
NameError: name 'df' is not defined
```

In []:

In []:

In []:

In []: